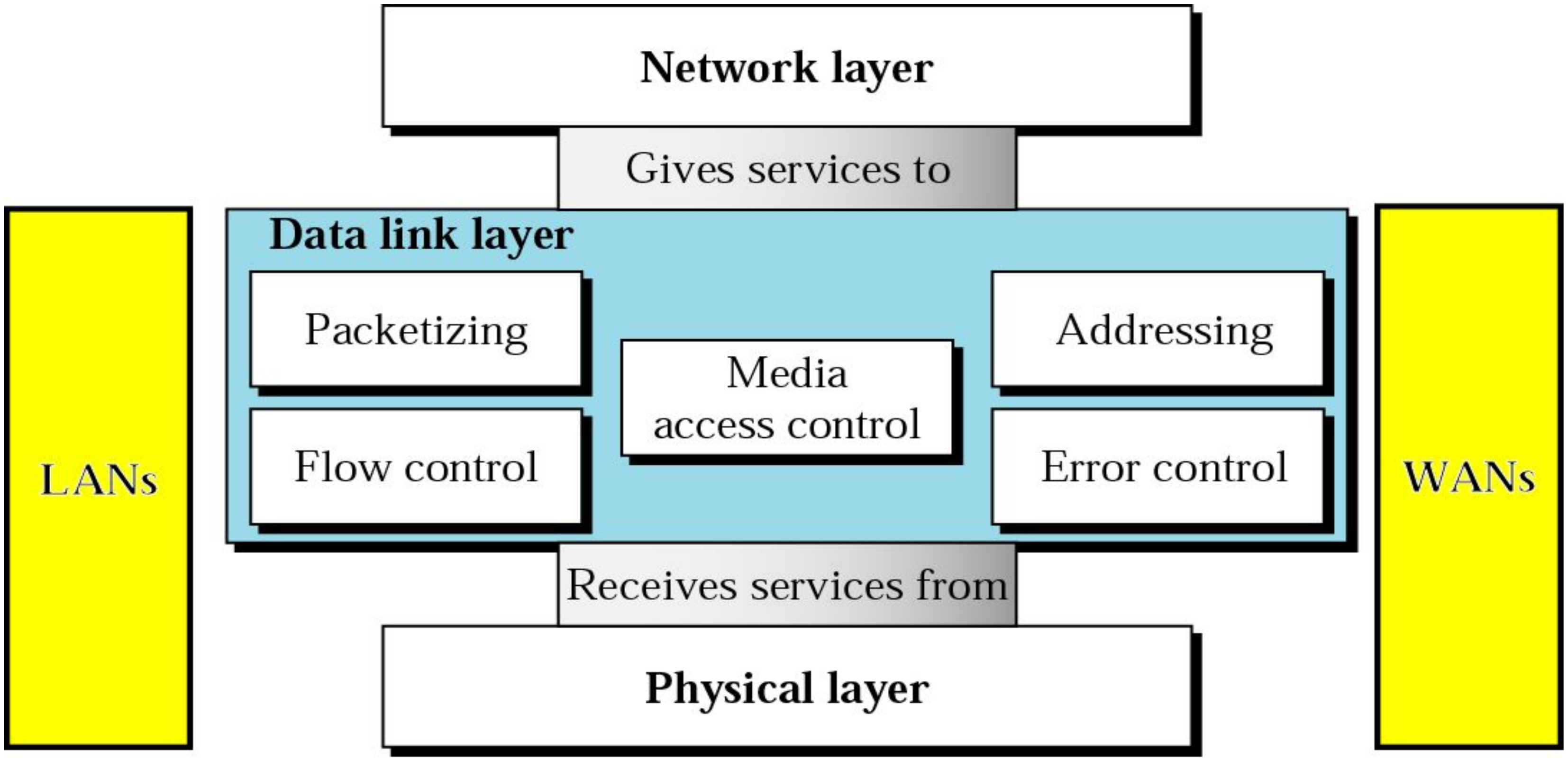
Computer Communication & Networks

Lecture 13

Datalink Layer: Datalink control and protocols

Data Link Layer



Data Link Layer Topics to Cover

Error Detection and Correction

Data Link Control and Protocols

Multiple Access

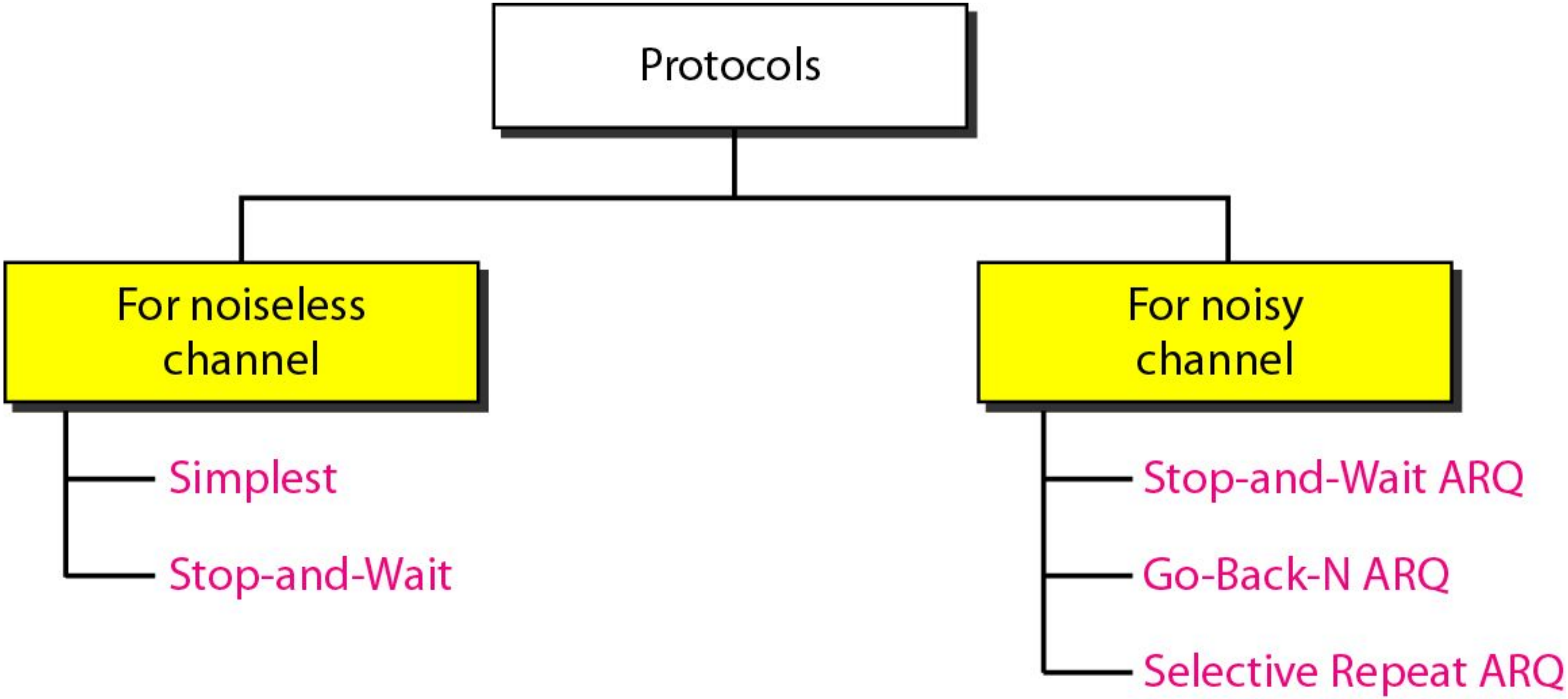
Local Area Networks

Wireless LANs

Flow & Error Control

- The most important responsibilities of the data link layer are flow control and error control. Collectively, these functions are known as data link control.
  - Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.
  - **Error control in the data link layer is based on automatic repeat request, which is the retransmission of data.**

Protocols (Contd.)



Noisy Channels

- We discuss three protocols in this section that use error control.
  - Stop & Wait Automatic Repeat Request
  - Go-Back-N Automatic Repeat Request
  - Selective Repeat Automatic Repeat Request

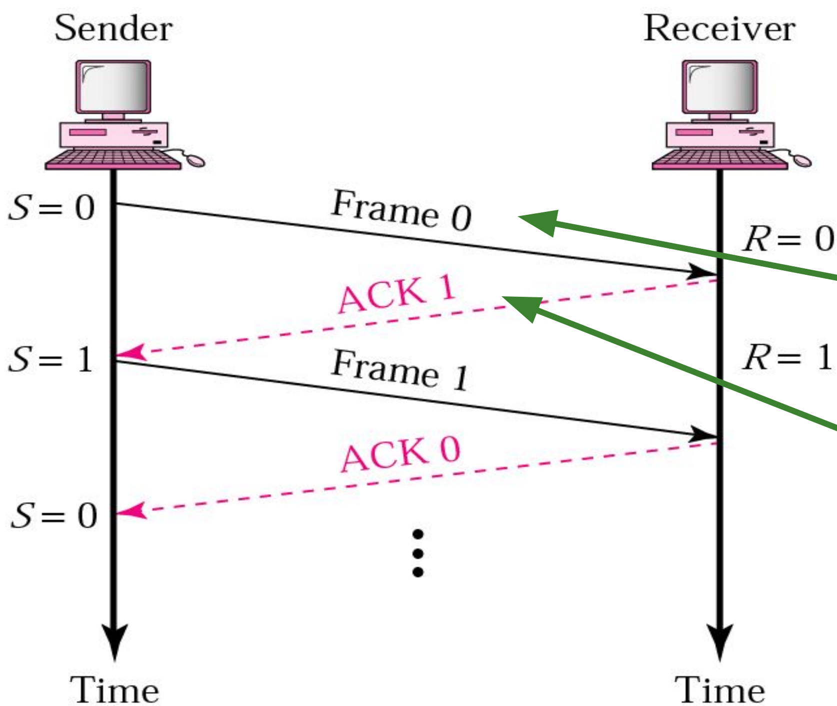
Stop-and-Wait Automatic Repeat Request

- Simplest flow and error control mechanism
- The sending device keeps a copy of the last frame transmitted until it receives an acknowledgement
  - identification of duplicate transmission (lost or delayed ACK)
- a damaged or lost frame is treated in the same way
- timers introduced
- positive ACK sent only for frames received safe & sound

A Simplex Stop-and-Wait ARQ

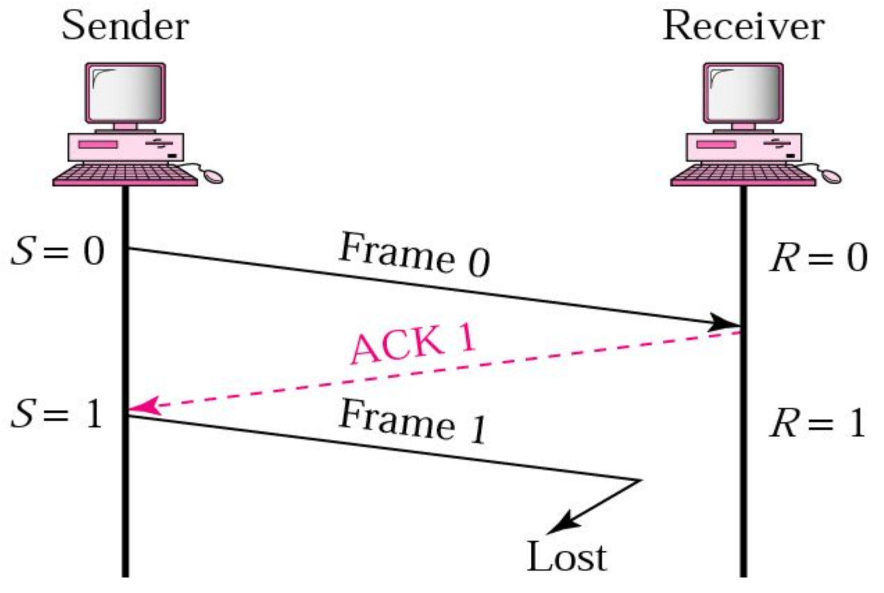
1. Normal operation
2. The frame is lost
3. The ACK is lost
4. The ACK is delayed

Stop-and-Wait ARQ
-Normal operation-



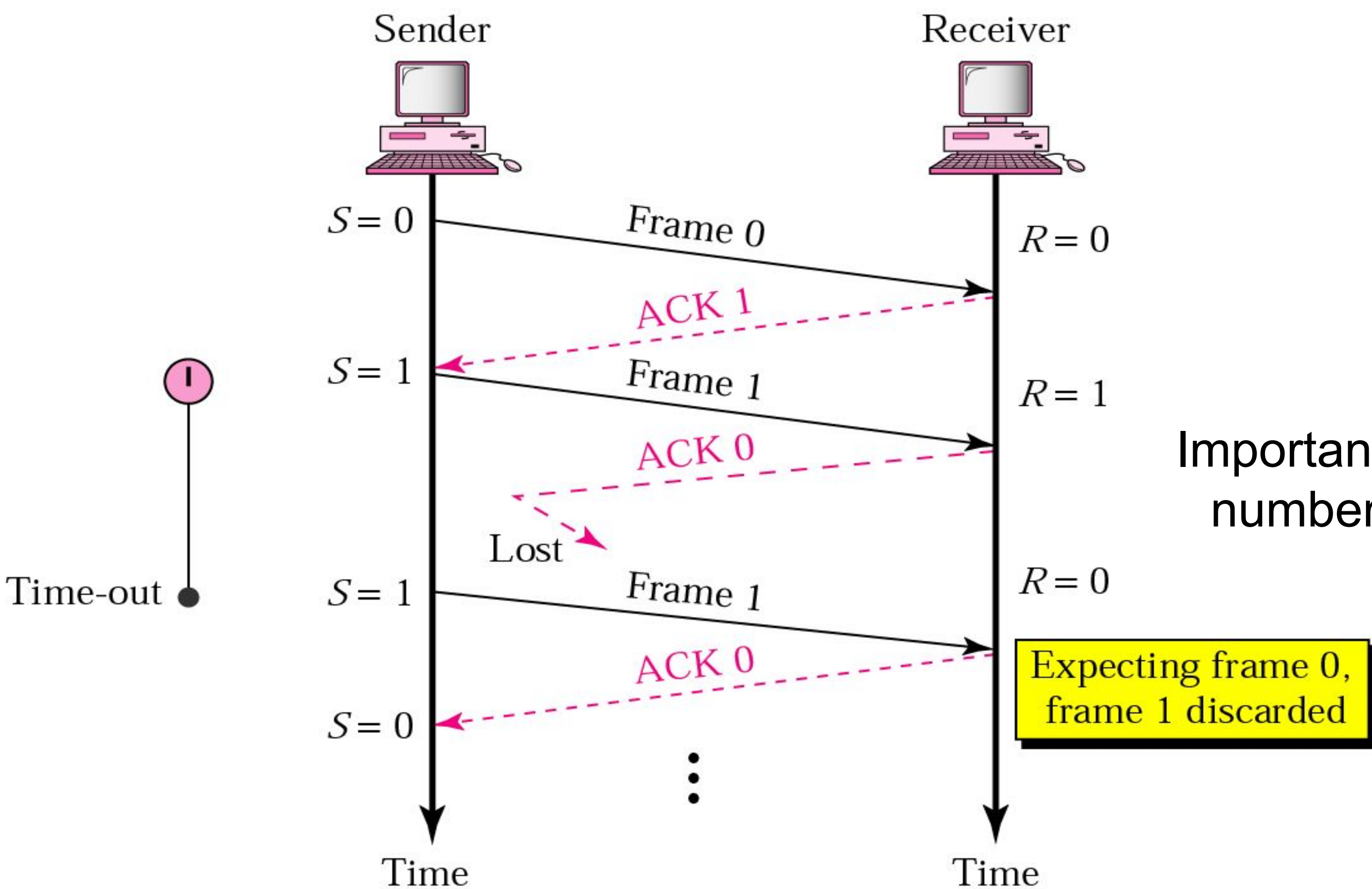
- The sender will not send the next piece of data until it is sure that the current one is correctly received
- sequence number necessary to check for duplicated packets
- No NACK – when packet is corrupted – duplicate ACKs instead

Stop-and-Wait ARQ
-Lost or damaged frame-



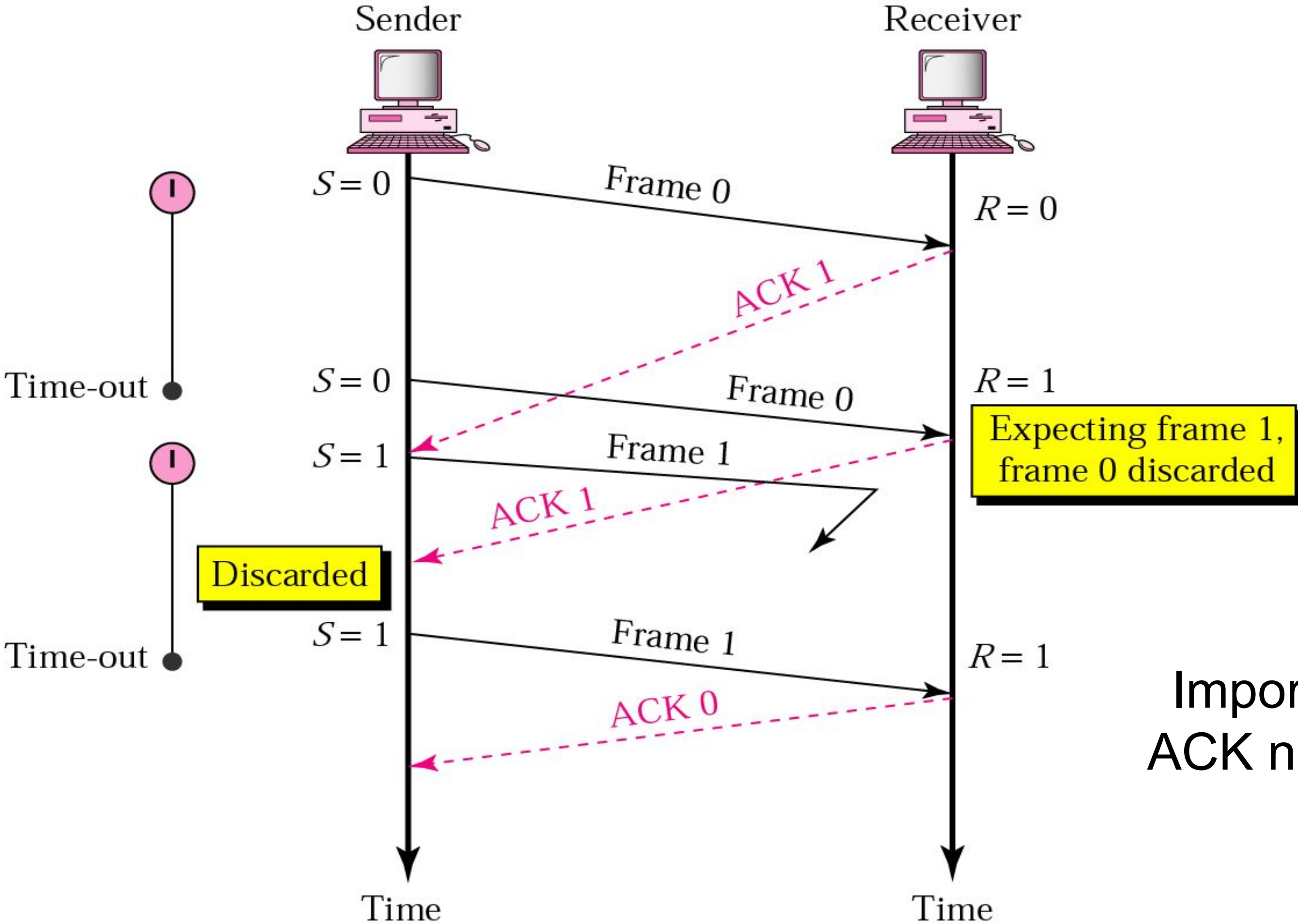
roundtrip delay
+
processing in the receiver

Stop-and-Wait ARQ
-Lost ACK-



Importance of numbering

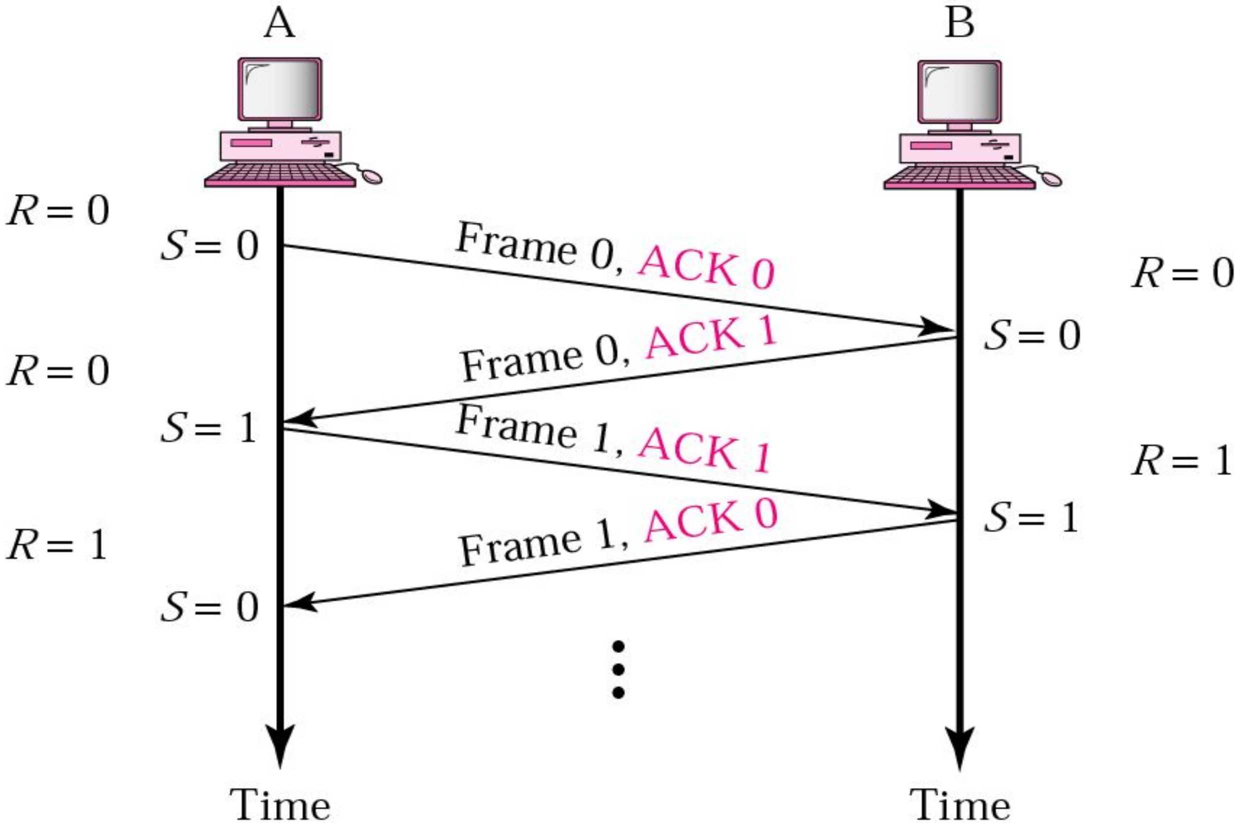
Stop-and-Wait ARQ
-Delayed ACK-



Importance of ACK numbering

Duplex Stop-and-Wait ARQ

- Piggybacking
  - combine data with ACK (less overhead saves bandwidth)



Drawbacks of Stop-and-Wait ARQ

- After each frame sent the host must wait for an ACK
  - inefficient use of bandwidth

- To improve efficiency ACK should be sent after multiple frames

- Alternatives: Sliding Window protocols
  1. Go-back-*N* ARQ
  2. Selective Repeat ARQ

Pipelining

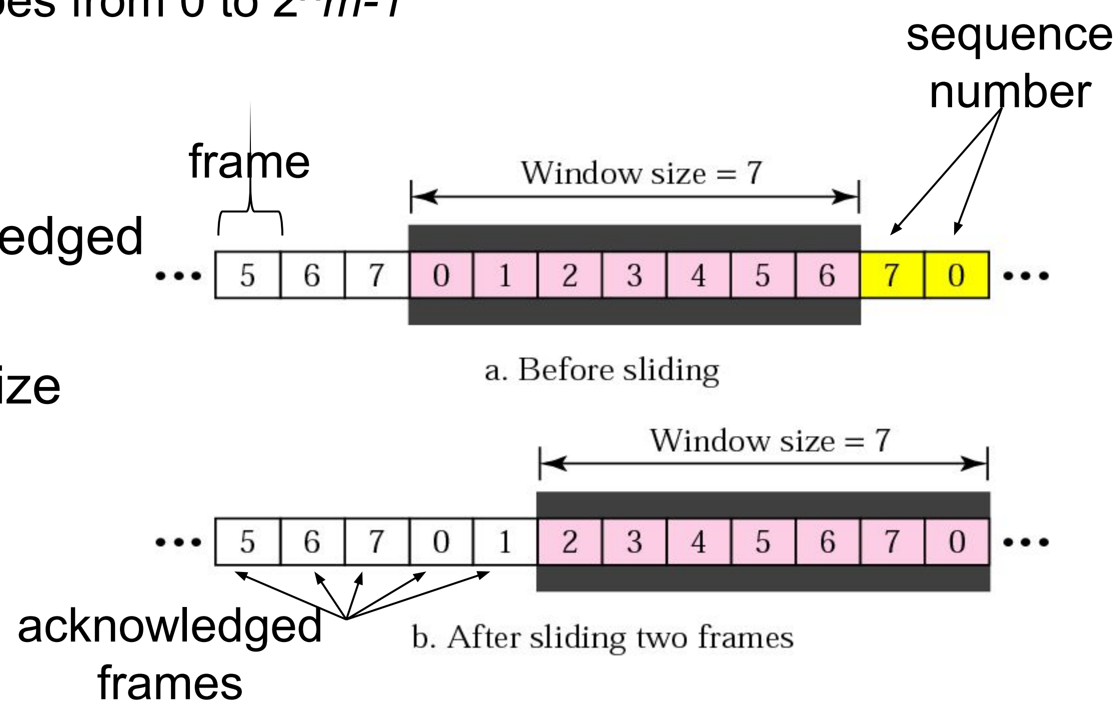
- One task begins before the other one ends
  - increases efficiency in transmission
- There is no pipelining in Stop-and-Wait ARQ

Sliding Window Protocols

- Sequence numbers
  - sent frames are numbered sequentially
  - number of frames stored in the header
    - if the number of bits in the header is *m* than <u>sequence number</u> goes from 0 to 2^*m-1*

- Sliding window
  - to hold the unacknowledged outstanding frames
  - the receiver window size

    always 1



Question

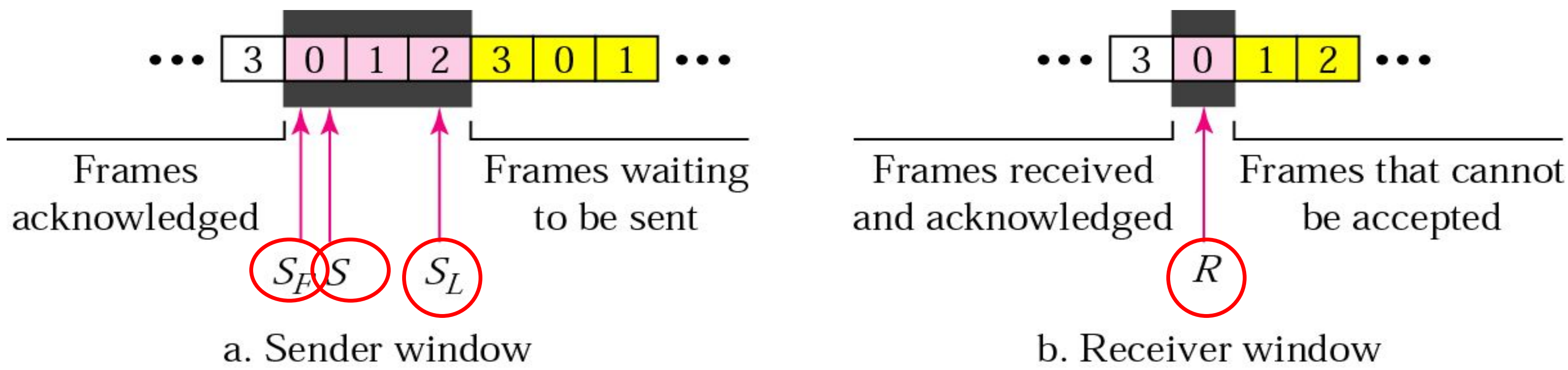
- What is the consequence of the window size in the receiver on the order of the received packets?

- Answer:

- Window size 1 means that the packets are received in order
  - Note: this is not the case for the larger receiver window size

Go-back-*N*
-Control variables-

- *S*- holds the sequence number of the recently sent frame
- *SF* – holds sequence number of the first frame in the window
- *SL* – holds the sequence number of the last frame
- *R* – sequence number of the frame expected to be received



The name of Go-back-*N*: why?

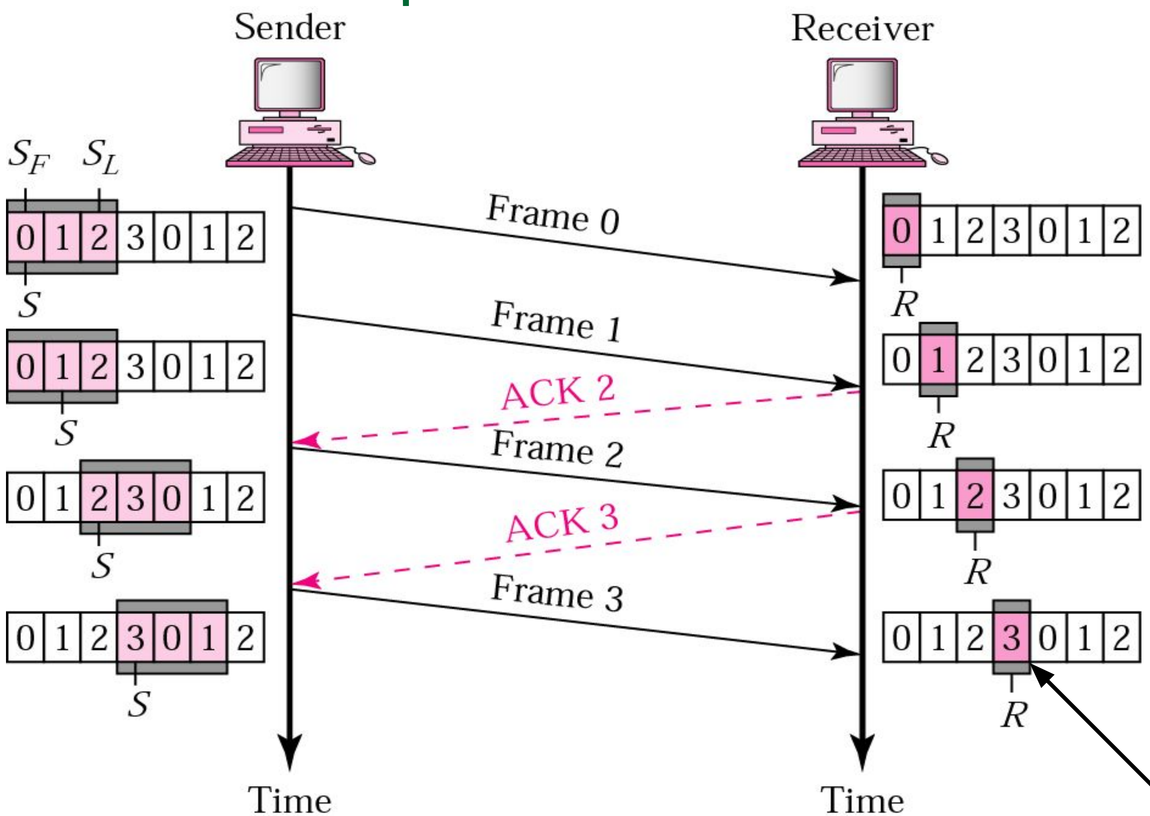
- Re-sending frame
  - when the frame is damaged the sender **goes back** and sends a set of frames starting from the last one ACKn'd
  - the number of retransmitted frames is ***N***

- **Example:**

- The window size is 4.
- A sender has sent frame 6 and the timer expires for frame 3 (frame 3 not ACKn'd). The sender goes back and re-sends the frames 3, 4, 5 and 6.

Go-back-N

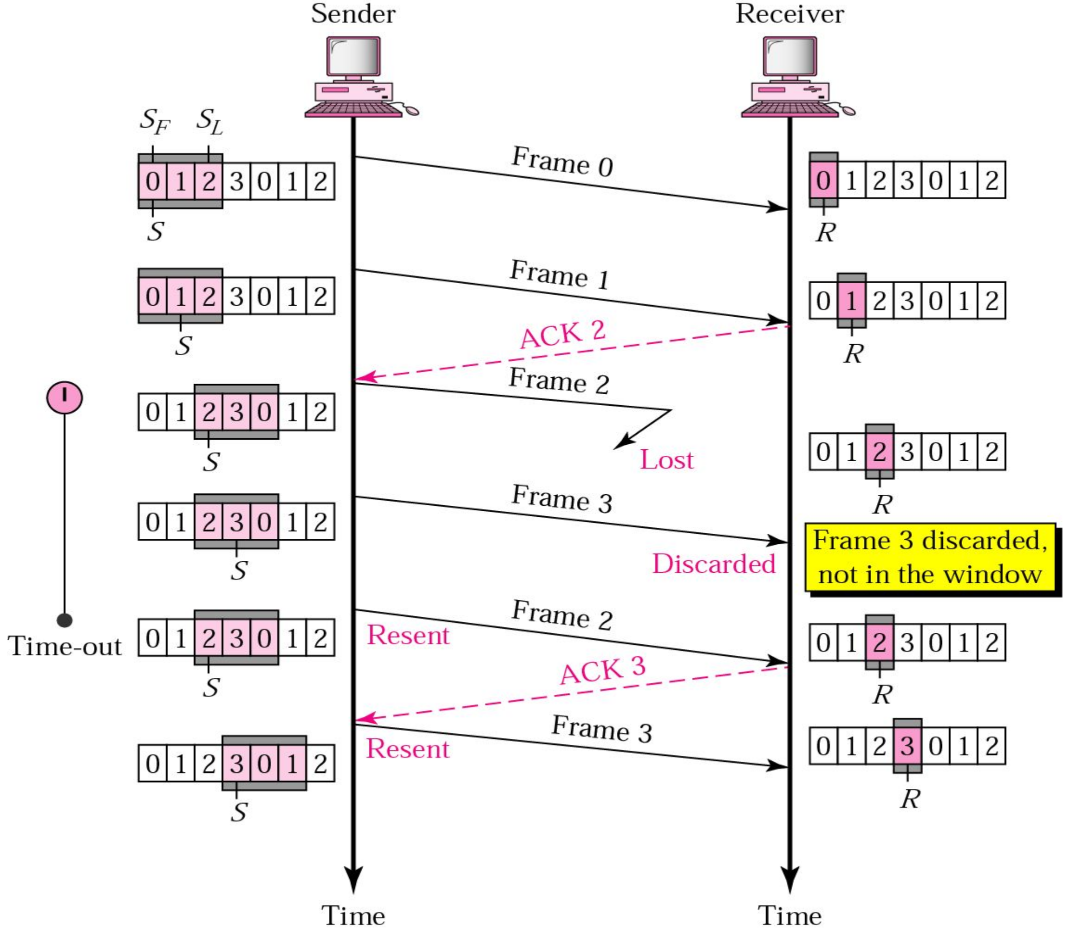
-normal operation-



- How many frames can be transmitted without acknowledgment?
- ACK1 – not necessary if ACK2 is sent
  - Cumulative ACK

expected sequence number

Go-back-*N*
-damaged or lost frame-



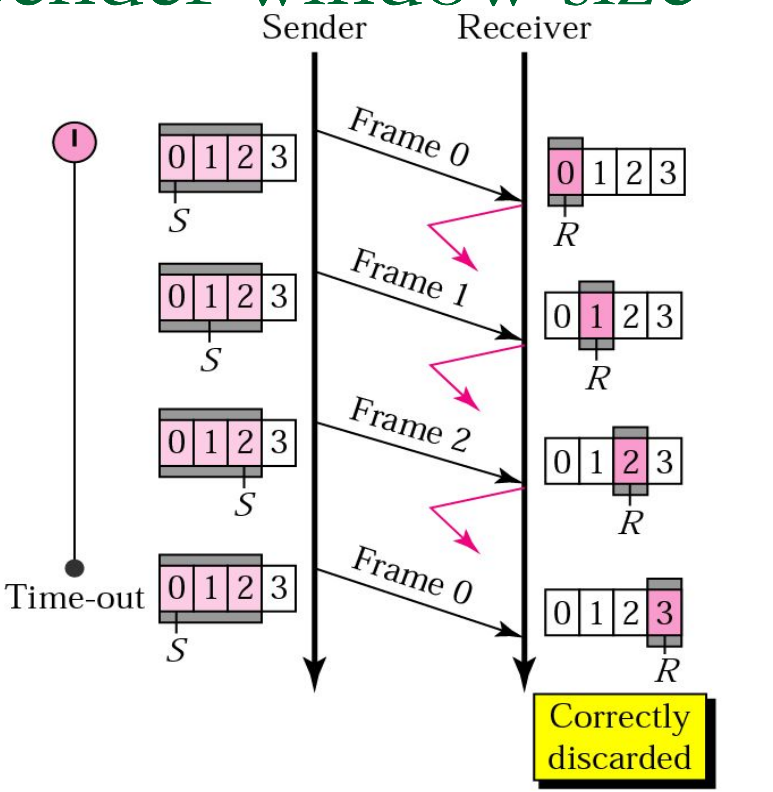
damaged frames are discarded!

Why are correctly received packets not buffered?

What is the disadvantage of this?

Go-back-*N*

-sender window size-



a. Window size < 2^m

Go-back-*N*

- Inefficient
  - all out of order received packets are discarded
- This is a problem in a noisy link
  - many frames must be retransmitted -> bandwidth consuming

- Solution
  - re-send only the damaged frames

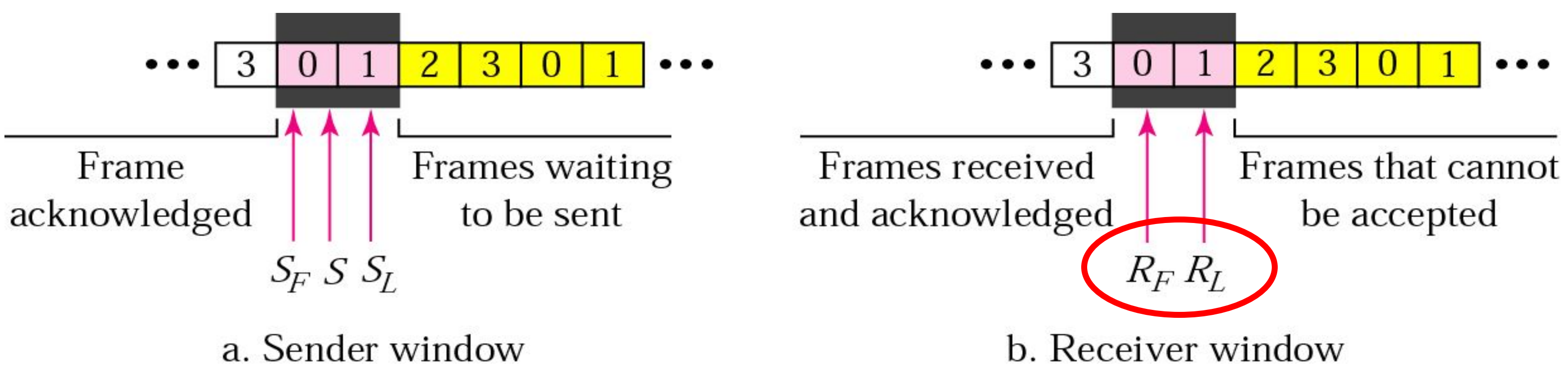
- Selective Repeat ARQ
  - avoid unnecessary retransmissions

Note

Stop-and-Wait ARQ is a special case of Go-Back-N ARQ in which the size of the send window is 1.

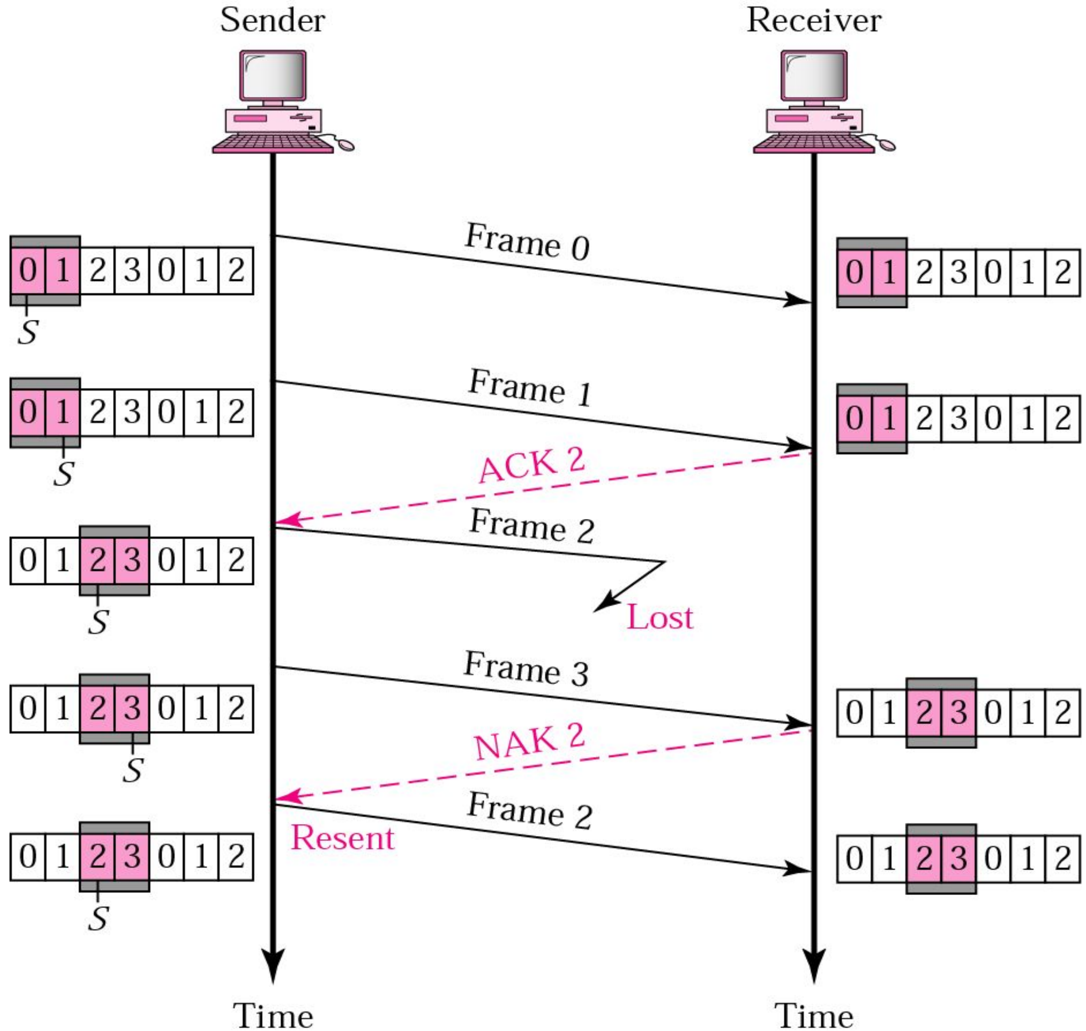
Selective Repeat ARQ

- Processing at the receiver more complex
- The window size is reduced to one half of 2^m
- Both the transmitter and the receiver have the same window size
- Receiver expects frames within the range of the sequence numbers

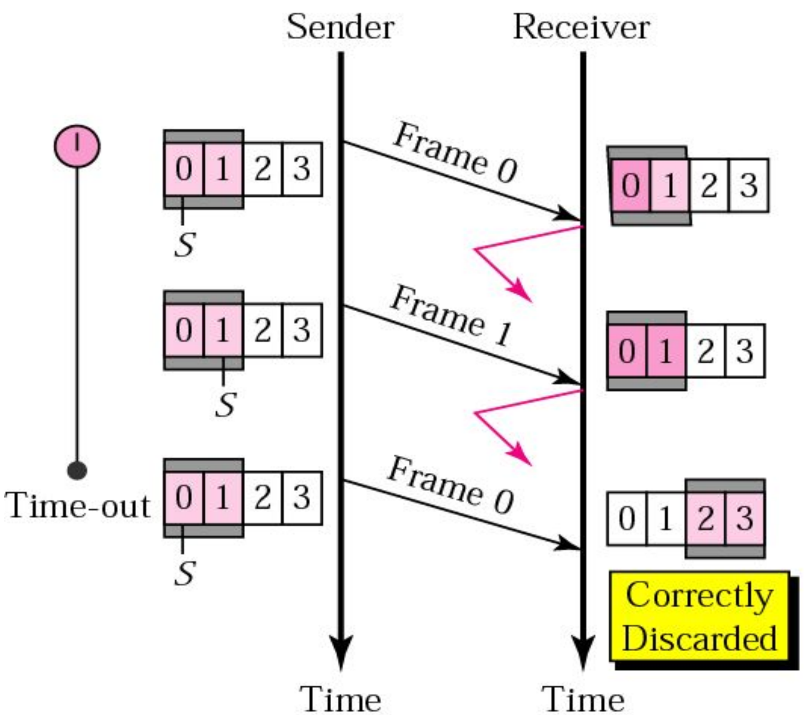


Selective Repeat ARQ
-lost frame-

Note:
retransmission triggered with NACK and not with expired timer



Selective Repeat ARQ
-sender window size-



a. Window size = 2^{m-1}

Readings

- Chapter 11 (B.A Forouzan)
  - Section 11.2,11.3, 11.5
  - (Cover only those contents which are related to topics covered in class)